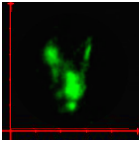


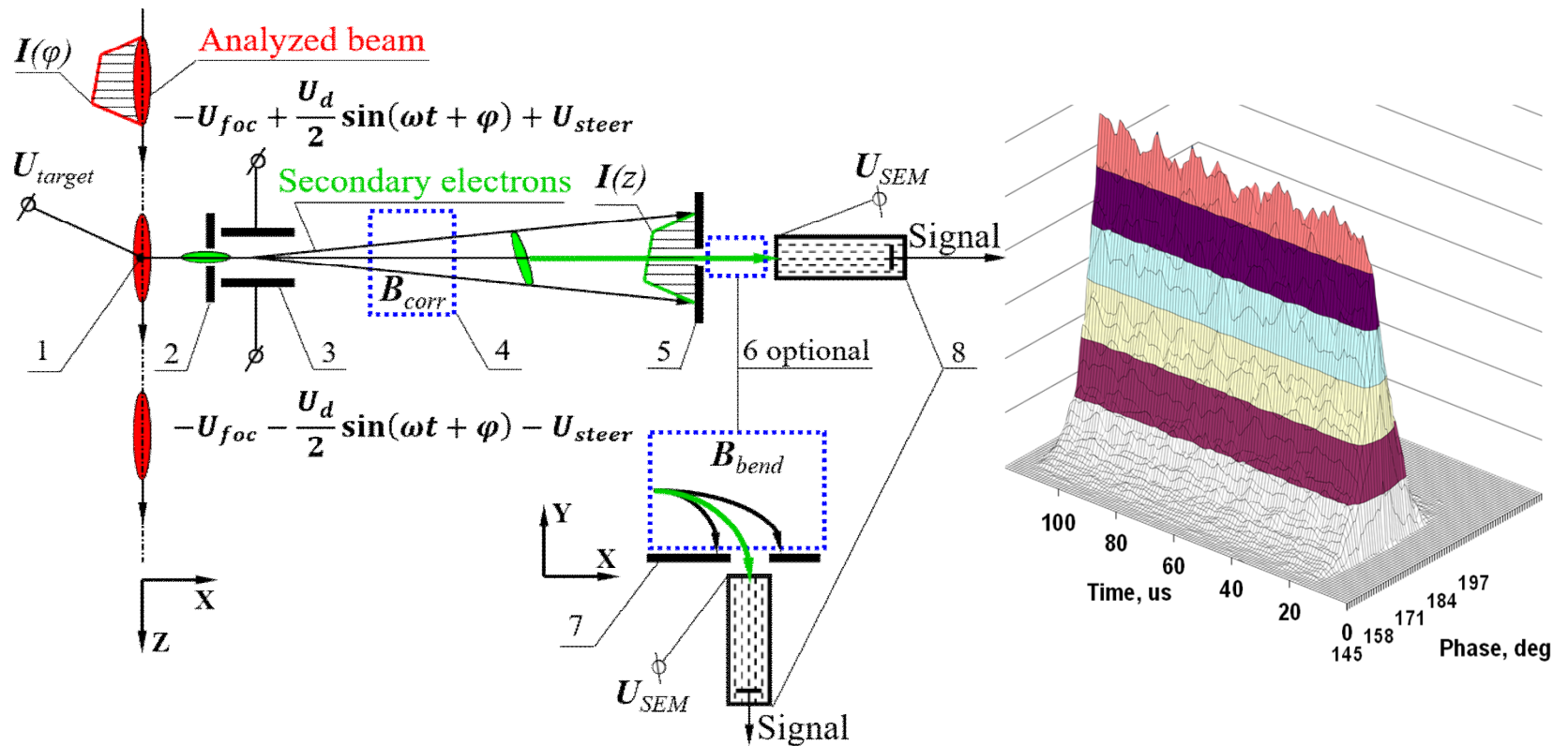
INR RAS instrumentation for bunch shape & beam cross-section monitoring

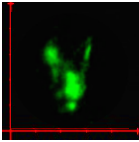
Sergei Gavrilov
Alexander Feschenko
Peter Reinhardt-Nickoulin
Accelerator Complex Division
Institute for Nuclear Research RAS



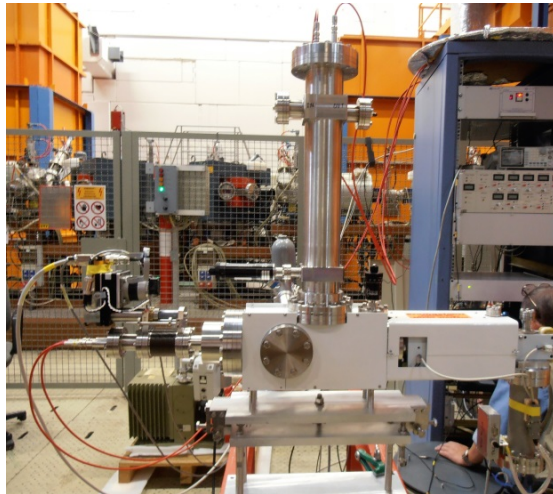


Bunch Shape Monitor: principles

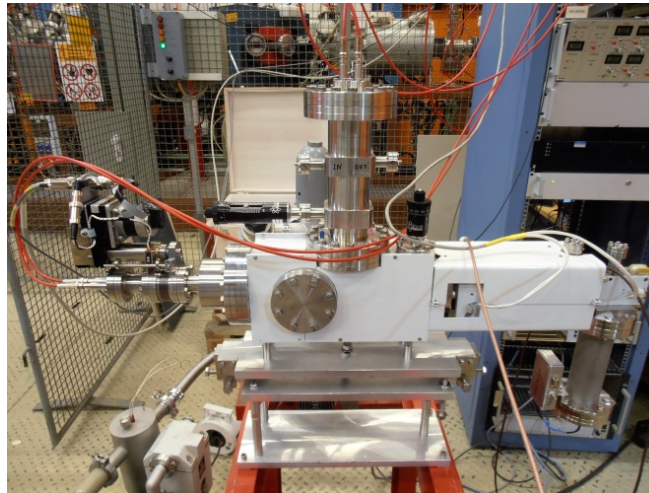
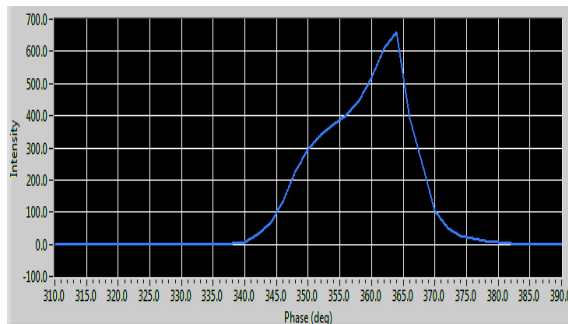




Bunch Shape Monitor: nowadays

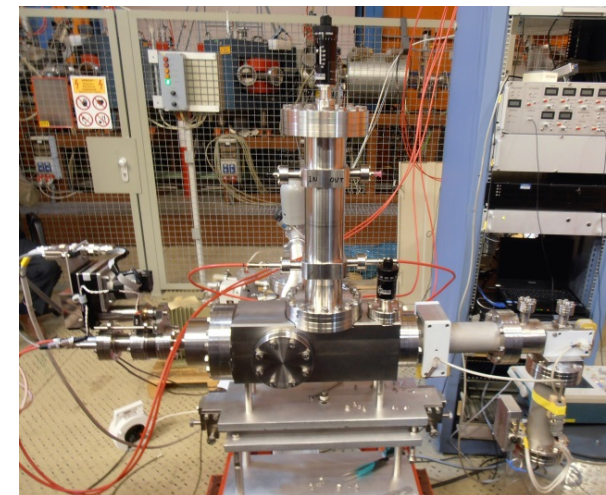


BSM UNILAC



BSM CW-linac

**360°
phase range**



BSM Proton linac

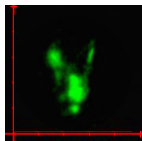


09.12.2016

«INR RAS instrumentation for BS & BCS monitoring»

Sergei Gavrilov 3/8





Bunch Shape Monitor: future

80.5 MHz

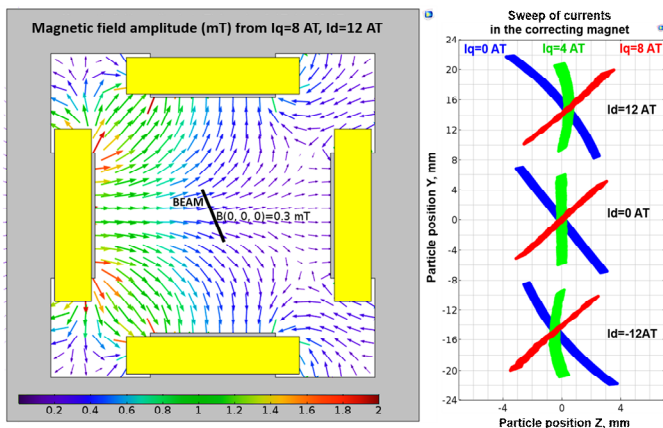
$$\Delta\varphi < 0.5^\circ$$



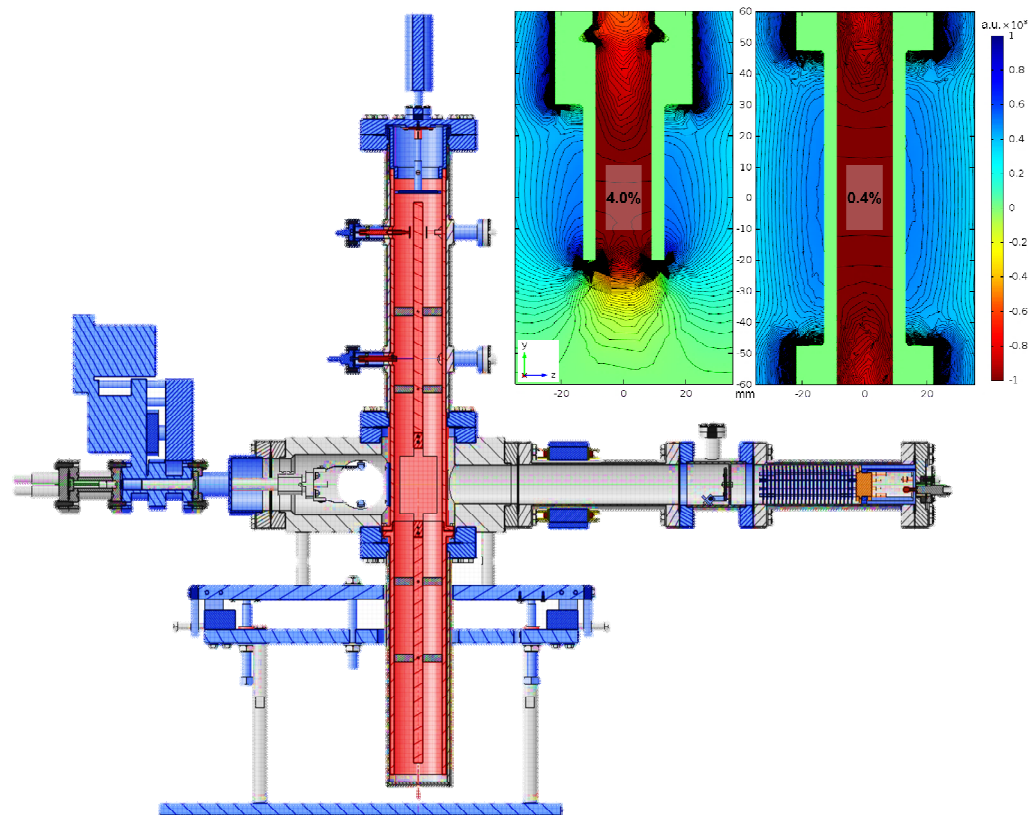
352.2 MHz

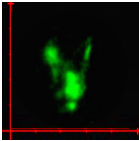
$$\Delta\varphi < 0.5^\circ$$

Double correcting magnet



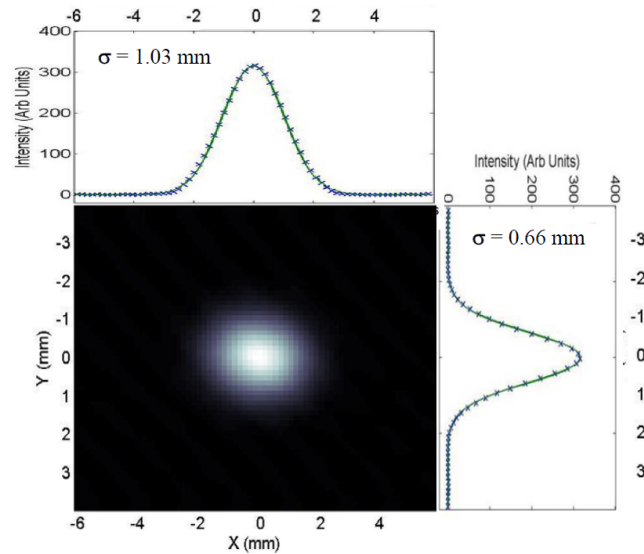
Symmetric RF-deflector



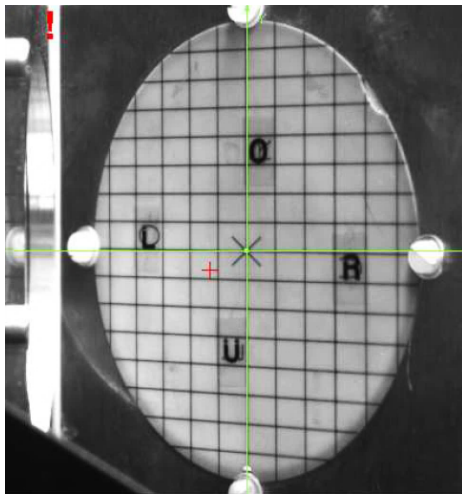


2D transparent transverse diagnostics at linacs: motivation

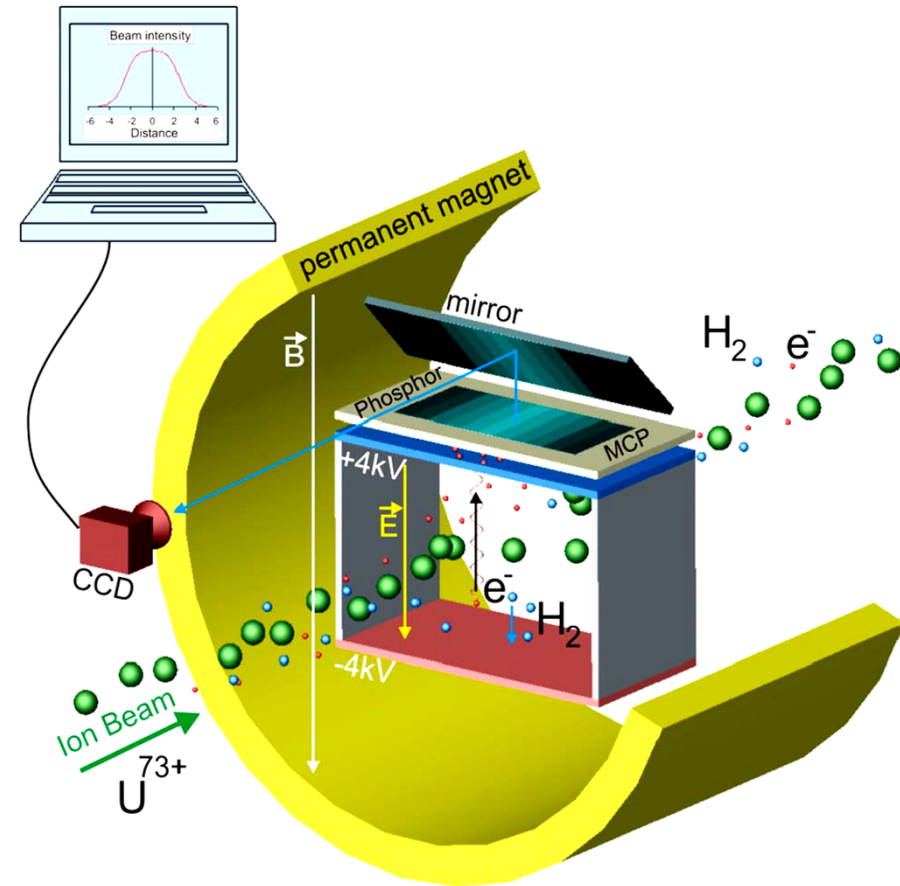
2D ...



... but destructive

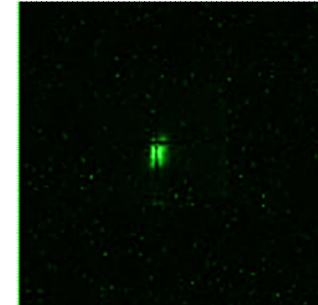
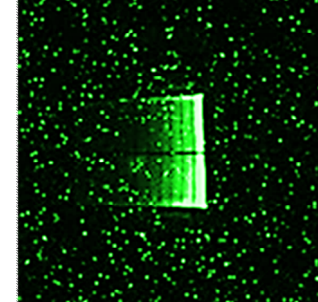
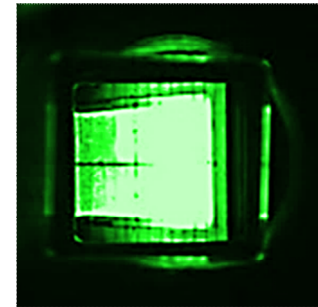
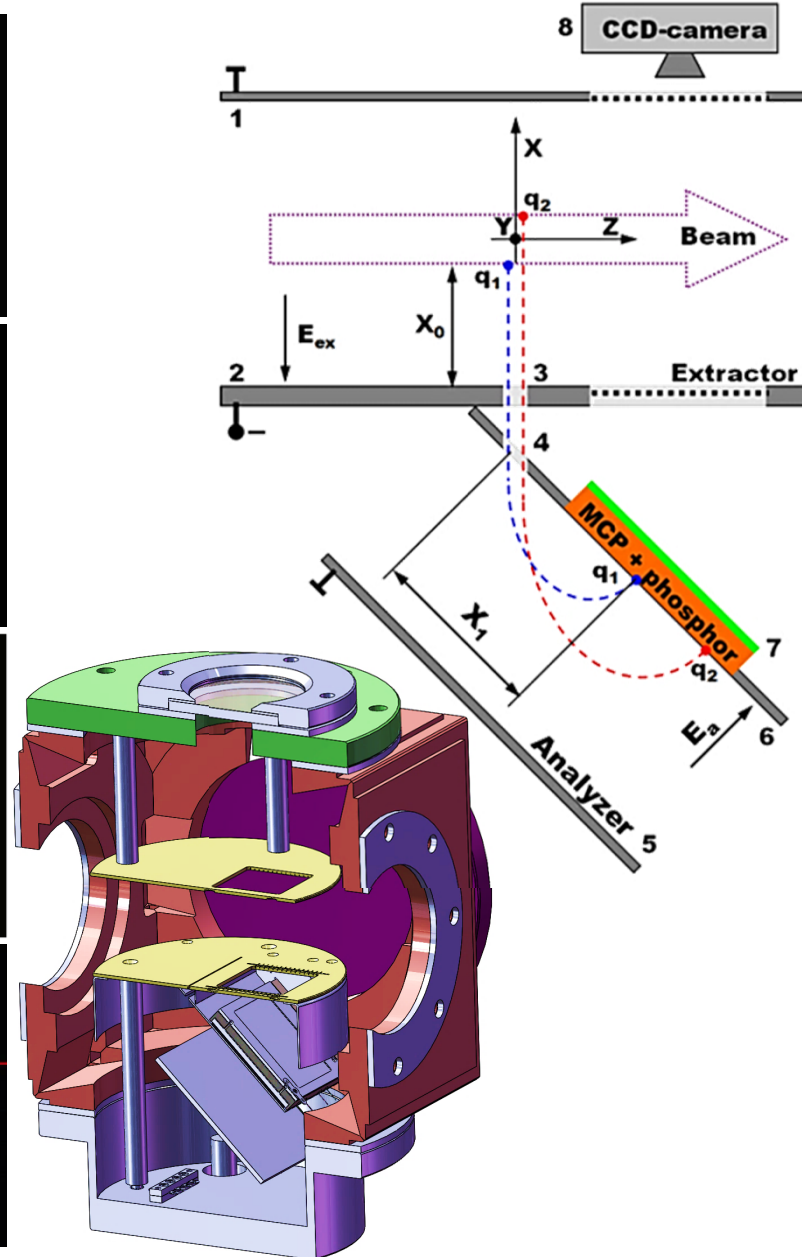
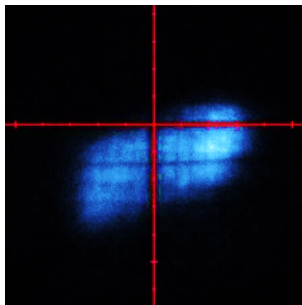
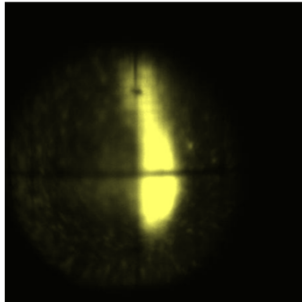
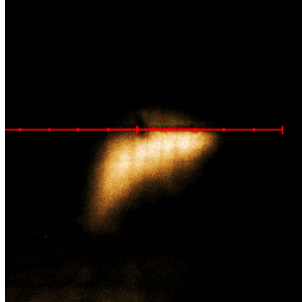
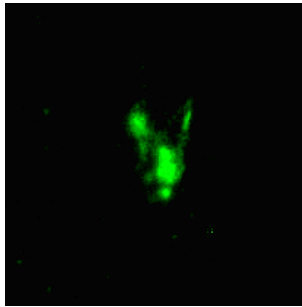
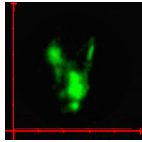


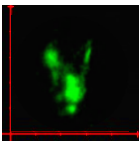
Transparent ...



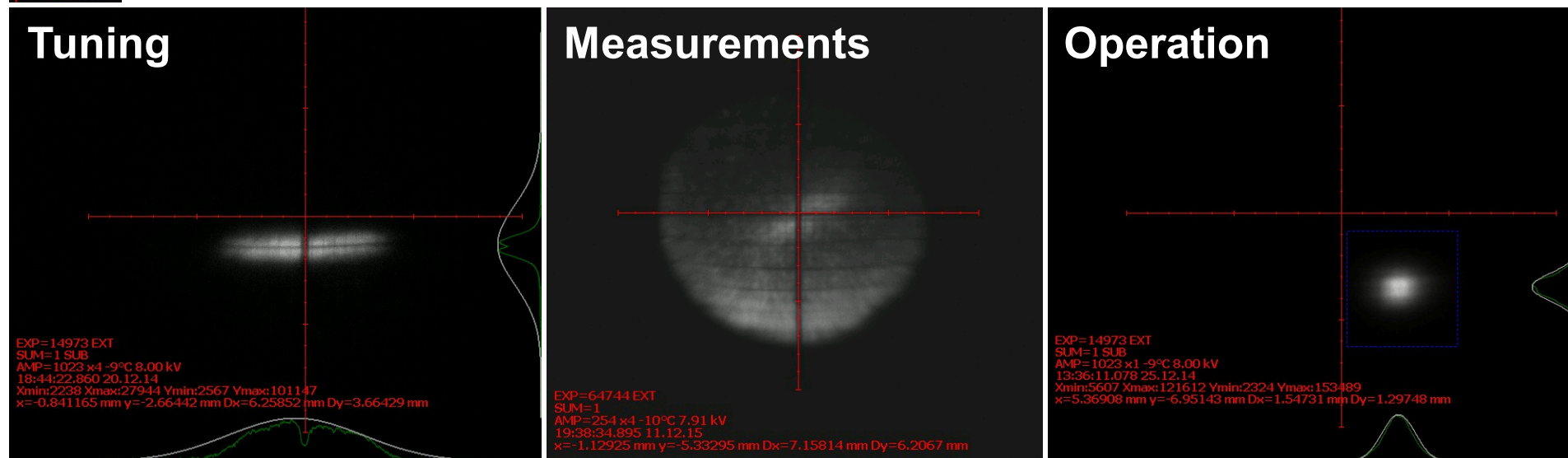
... but 1D

2D & transparent: **B**eam **C**ross-**S**ection **M**onitor

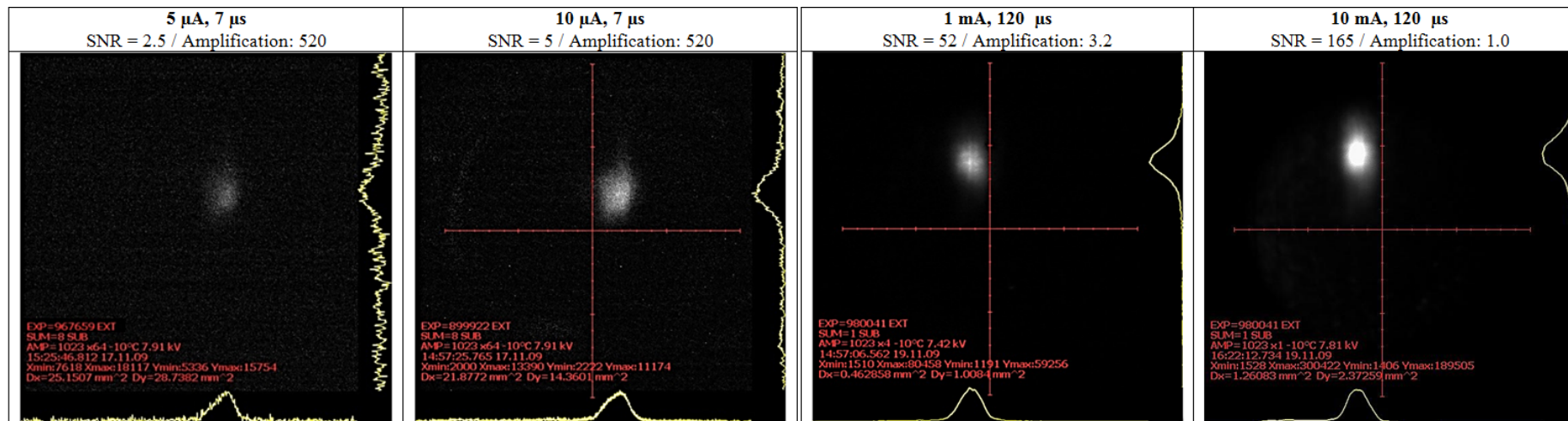


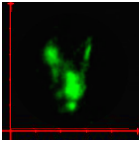


Beam Cross-Section Monitor: results



Measuring range

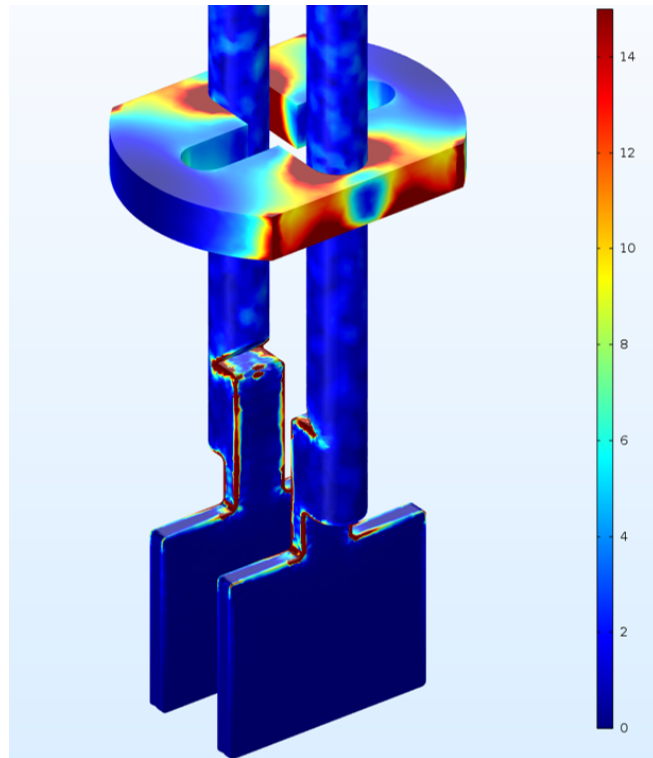




Extra reports today

THPSC055

Electrodynamic Characteristics of RF-Deflector for Bunch Shape Monitor



THPSC058

Beam diagnostics and instrumentation upgrade for Multipurpose Research Complex of INR RAS

